

FIT1055 IT Professional Practice and Ethics Revision notes

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Week 2 Revision Notes: Effective Communication & Oral Presentations

Key Learning Outcomes

By the end of this week, you should be able to:

- Understand communication theory and its relevance to IT professionals
 - Distinguish between intrapersonal and interpersonal communication skills
 - Apply effective communication techniques (assertiveness, listening, questioning, feedback)
 - Use the Minto Pyramid Principle for presentations
 - Demonstrate effective oral and written communication strategies
-

1. THEORY OF COMMUNICATION

Speech Act Theory (Austin & Searle)

Key Concept: Utterances don't just convey meaning—they perform actions.

Performative Utterance: Words that simultaneously express AND do something.

Example Scenarios:

- "I am hungry" = Expressing hunger + Requesting food
- "You're fired!" = Stating employment status + Ending employment
- "Would it be too much trouble to hand me that wrench?" = Asking two questions (capability + actual request)

Quiz Tip: Understand that language accomplishes actions, not just information transfer.

Common Ground (Herbert H. Clark)

Key Concept: Effective communication requires establishing mutual understanding.

Grounding Process Requires:

1. Listener notices something was said

2. Listener hears what was said
3. Listener understands what was said
4. Listener understands what was meant

Common Ground Includes:

- Common cultural and social history
- Public history of the interaction
- Current public state of the interaction

Application Example: In a team project meeting, before diving into technical details, ensure everyone understands the project context, their roles, and previous decisions made.

2. PROFESSIONAL COMMUNICATION

Intrapersonal vs Interpersonal Skills

Intrapersonal	Interpersonal
Communication with oneself	Communication with others
Self-talk, reflection	Verbal/non-verbal exchange
Skills needed: Self-discipline, open-mindedness, ability to learn, self-motivation, patience, positive attitude	Skills needed: Assertiveness, active listening, questioning, feedback
Importance: Knowing your strengths, limitations, what drives/frustrates you	Importance: Building relationships, teamwork, organizational productivity

Quiz Scenario: If asked about preparing for a difficult conversation with your manager, you'd use intrapersonal skills (self-awareness, managing emotions) PLUS interpersonal skills (assertiveness, active listening).

3. EFFECTIVE COMMUNICATION TECHNIQUES

A. ASSERTIVENESS TECHNIQUES

Definition: Obtaining desired outcomes without contravening others' rights.

The 9 Assertiveness Skills:

1. **Workable Compromise**
 - Used when others become frustrated
 - Finding middle ground to satisfy both parties

- *Example:* Team member wants feature X, you prefer Y → Negotiate feature with elements of both
 - 2. **Say No**
 - Simple refusal followed by explanation
 - *Example:* "No, I can't take on that task this week because I'm already committed to the deployment deadline"
 - 3. **Dismiss and Redirect**
 - Address unimportant points, then redirect to what matters
 - *Example:* "I understand the font choice concerns you, but let's focus on the data accuracy first"
 - 4. **Fogging**
 - Calmly acknowledge criticism without becoming defensive
 - Takes force out of criticism
 - *Example:* "You're right that the report was late. Let me explain the technical issues we encountered"
 - 5. **Forcing a Choice**
 - Clarify priorities when overloaded with tasks
 - *Example:* "I can complete either the testing or the documentation by Friday. Which is the priority?"
 - 6. **Broken Record**
 - Calmly repeat your message without frustration
 - Used when message isn't received first time
 - *Example:* Repeatedly asking for required access permissions: "I still need database access to complete this task"
 - 7. **Ask for Specifics**
 - Request concrete examples when facing vague criticism
 - *Example:* "Can you give me specific examples of where my code documentation is unclear?"
 - 8. **Questioning to Prompt Awareness**
 - Challenge thinking through questions
 - Use carefully—know how person will respond
 - *Example:* "How do you think this architecture decision will affect system scalability?"
 - 9. **Threat**
 - Last resort to highlight consequences
 - Always offer "carrot" with "stick"
 - *Example:* "If we don't address this security vulnerability now, we'll face audit failure, but if we allocate resources this week, we can prevent it"
-

B. LISTENING SKILLS

Two Key Techniques:

1. **Non-verbal Communication**
 - Listen to words

- Observe non-verbal behavior
 - Evaluate if content matches behavior
 - *Example:* Colleague says "I'm fine with the deadline" but looks stressed → Ask follow-up questions
2. **Listening Responsiveness**
- Show attentiveness through:
 - Nodding
 - Upright, forward-leaning posture
 - Orienting body toward speaker
 - Eye contact
 - Facial expressions (smiling/appropriate reactions)

Quiz Tip: Both techniques work together to show engagement and gather complete information.

C. ACTIVE LISTENING

Definition: Using minimal verbal responses to encourage speaker to articulate and clarify meaning.

Key Techniques:

- Clarifying the speaker's meaning
- Checking accuracy of what was said
- Summarizing back what they said
- Acknowledging without committing

Application Example:

- Team member: "The database is slow"
- Active listening response: "So you're saying query performance has degraded? Can you clarify which specific operations are affected?"

D. QUESTIONING TECHNIQUES

Three Main Types:

1. **Closed Questions**
 - Yes/no answers
 - Establish facts, get commitments
 - *Example:* "Have you completed the unit tests?"
2. **Open Questions**
 - Broad scope of response
 - Use: who, what, when, where, how
 - *Example:* "What challenges did you encounter during integration testing?"

3. Direct Probe

- Specific, direct question
- Encourage deep thought
- *Example:* "How will this design pattern improve maintainability?"

Other Question Types:

- **Objective Criteria:** Focus on facts to defuse contentious situations
 - **Testing:** Check knowledge/skills or build rapport
 - **Hypothetical:** Explore options ("What if...?")
 - **Reflective:** Reflect feelings back to show acknowledgment
-

E. FEEDBACK SKILLS

Good Feedback Must Be:

- Fair
- Accurate
- Specific
- Formally structured
- Solution-oriented
- Focused on behavior, not personality

Application Example: ✗ Poor: "Your code is messy" ✓ Good: "I noticed the authentication module has inconsistent naming conventions. Using camelCase throughout would improve readability. Would you like me to share our coding standards document?"

4. EFFECTIVE WRITTEN COMMUNICATION

5-Step Approach:

1. **Identify Objectives**
 - What am I writing for? (Job application, proposal, request)
2. **Conduct Research**
 - What does the audience need? What skills/info do they seek?
3. **Create Background**
 - Who am I? How did I find this opportunity? Why should they care?
4. **Compose Message**
 - Clear statement of who's writing to whom
 - Clarify purpose
 - Provide details (clear, concise, conclusive)
 - Create initiative for readers
 - End nicely

5. Follow-Up

- Determine appropriate follow-up
- Assess consequences

Professional Email Example Structure:

Dear Mr. Sato,

[CLARIFICATION] My name is Suzie, and I am writing to ask whether your lecture slides for FIT2035 can be made available before each lecture.

[DETAILS] Last year, I found the lecture slides extremely useful in preparation for lecture attendance.

[CONSIDERATION + INITIATIVE] I understand slides can't always be made available prior, but I would be grateful if you could make them available during the week.

[CONCLUSION] Thank you,
Suzie

5. ORAL PRESENTATION SKILLS

The 4-Step Strategy:

1. PLANNING

- Define topic and goal
- Analyze audience (size, characteristics, needs)
- Consider logistics (room, equipment, time)

Quiz Question Type: "What should you consider when planning a presentation for senior management vs. technical team?"

2. DEVELOPMENT

- Choose presentation type: Informational, Motivational, Situational
- Outline key points for introduction, body, conclusion
- Determine supporting information needed

3. WRITING (NARRATIVES)

Introduction Must Cover:

- Context
- Aim
- Structure

Conclusion Must Cover:

- Summary
- Key takeaway points
- Closing remarks

Example: "We will be speaking on the article titled 'Cloud Security Challenges,' written by security expert Jane Doe. In this presentation, we will provide the main highlights. I will introduce the context, then X will discuss key findings, followed by Y who will conclude with recommendations."

4. DELIVERY

Remember: Who presents matters!

6. VISUAL AIDS & MINTO PYRAMID PRINCIPLE

Why Use Visual Aids?

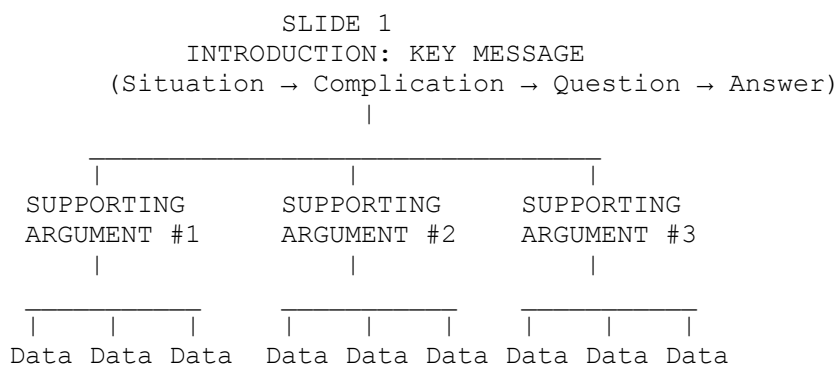
- Improves understanding and memory
- Serves as notes
- Provides clear organization
- Facilitates eye contact and movement
- Contributes to credibility

10-20-30 Rule:

- **10** slides maximum
- **20** minutes maximum speaking time
- **30** point minimum font size

MINTO PYRAMID PRINCIPLE

Structure:



Key Principle: Start with the answer/main message, then support with logical arguments, each backed by data.

Application Example for IT Project Presentation:

Slide 1: "We should migrate to microservices architecture"

- **Supporting Argument 1:** Improved scalability (Slides 2-4 with data)
- **Supporting Argument 2:** Enhanced maintainability (Slides 5-7 with data)
- **Supporting Argument 3:** Better fault isolation (Slides 8-10 with data)
- **Conclusion:** Summary and action items

QUIZ PREPARATION CHECKLIST

Concepts to Master:

- ☐ Difference between performative utterance and regular statement
- ☐ Four steps of grounding in conversation
- ☐ Intrapersonal vs interpersonal skills and examples
- ☐ All 9 assertiveness techniques with scenarios
- ☐ Difference between listening, active listening, and responsiveness
- ☐ When to use open vs closed questions
- ☐ Six characteristics of good feedback
- ☐ 5-step approach to professional writing
- ☐ 4-step presentation strategy
- ☐ Minto Pyramid structure

Practice Scenarios:

1. Identify which assertiveness technique to use in given situations
2. Write professional emails using correct structure
3. Identify good vs poor feedback examples
4. Match question types to appropriate contexts
5. Organize presentation content using Minto Pyramid

Common Quiz Question Formats:

- "Which assertiveness technique is being demonstrated?"
- "Identify the missing element in this communication"
- "What type of question would be most appropriate here?"
- "Arrange these slides according to Minto Pyramid Principle"
- "What makes this feedback effective/ineffective?"

Final Tip: Communication is about building COMMON GROUND. Every technique learned this week serves that ultimate goal—whether writing, speaking, listening, or presenting, you're working to ensure mutual understanding and effective action.

Week 3 Revision Notes: Professional Research Skills

1. Core Research Concepts

What is Research?

Investigation of materials, sources, or subjects to establish facts and contribute to knowledge.

Research Skills Definition

Transferable soft skills used to discover ideas, analyze information from various sources, and solve problems effectively.

Why Important for IT Careers?

- Highly sought after by employers
- Transferable across roles and sectors
- Essential for professional growth and adaptation

Key Transferrable Skills Developed

- Time management
- Clear communication
- Problem-solving
- Critical thinking
- Initiative taking

2. Research in the Age of Generative AI

Critical Insight

With AI handling routine tasks, the ability to **evaluate AI outputs** and make **higher-order decisions** is crucial.

Key Research Finding

Higher confidence in GenAI is associated with **less critical thinking**, while higher self-confidence is associated with **more critical thinking**.

Main Barriers to Critical Thinking with AI

- Lack of time for critical thinking
 - Not possessing enough domain knowledge
-

3. The Intelligent Research Cycle Process

Definition: A process by which data and information can be meaningfully converted into "intelligence" that benefits the organization or people.

The Complete Cycle (Know This Sequence!)

1. **Direction and Planning**
 2. **Collection of Data and Information**
 3. **Processing of Data and Information**
 4. **Analysis** (includes 3 sub-processes)
 - Evaluating sources
 - Integrating sources
 - Acknowledging sources
 5. **Dissemination**
 6. **Feedback** (loops back to start)
-

4. Stage-by-Stage Breakdown

Stage 1: Direction and Planning

What to do:

- Define specific research question
- Set clear objectives
- Set scope and outcome
- Create timeline with deadlines
- Identify needed source types

Example Application:

Scenario: Research ethical consequences of ChatGPT in doctor training

Research Question: "How does Generative AI impact patient safety?"

Objectives:

- Identify common ethical issues
- Understand strengths and limitations
- Assess real-world impacts

Scope: Different OpenAI techniques in healthcare

Outcome: Concise report with pros and cons

Stage 2: Collection/Gathering of Data

What to do:

- Develop collection/gathering plan
- Map available sources to requirements
- Search credible sources (Google Scholar, IEEE Xplore, ACM Digital Library)
- Use search operators effectively
- Collect diverse perspectives (technical, legal, social, ethical)

Search Strategies:

- **Google Advanced Search:** Use operators to refine results
 - **Monash Library:** Use Boolean operators (AND, OR, NOT)
 - **Gen AI:** Use specific prompts for source discovery
-

Stage 3: Processing of Data and Information

What to do:

- Read systematically with research question in mind
- Extract key information relating to objectives
- Organize by themes/categories
- Create focused summaries

Key Point: This is **systematic analysis**, not casual reading!

Stage 4: Analysis (Three Sub-Processes)

4a. Evaluating Sources

Use the CAAS Framework:

Criterion	Question to Ask
Currency	How up-to-date is the information?
Accuracy	Is the information correct? Can I verify it?
Authority	Does the author/organization have expertise?
Accessibility	How easy is it to find/understand?
Stability	Is the information stable over time?

Example Evaluation:

Source: "The Ethics of ChatGPT in Medicine" (Haltaufderheide & Ranisch, 2024)

- ✓ Currency: Published 2024 (very current)
- ✓ Accuracy: References from first-tier journals
- ✓ Authority: Authors have expertise in AI ethics in medicine
- ✓ Accessibility: Open journal, freely accessible
- ✓ Stability: Peer-reviewed review paper (stable)

Conclusion: HIGH QUALITY SOURCE

4b. Integrating Sources

Purpose: Define problem statements using How-What-Why framework

Pattern Identification Process:

1. Identify recurring themes across sources
2. Note agreements and contradictions
3. Look for connections between perspectives
4. Synthesize into coherent understanding

How-What-Why Framework:

Component	Question	Example from Healthcare AI
How	How should AI ideally work?	AI should integrate into EHR systems seamlessly, enhance workflow without replacement
What	What ethical issues arose?	Hallucinations, bias, discrimination, erosion of patient autonomy
Why	Why did these issues occur?	Algorithmic reductionism, lack of patient involvement, inadequate oversight

Application Example:

Research Topic: ChatGPT in Healthcare

1. Identify Patterns (How it should work):

- Medical notes creation and summarization
 - Patient monitoring interpretation
 - Diagnostic search assistance
2. Identify Issues (What went wrong):
- Pattern 1: Accuracy Issues
- Hallucinations in critical contexts
 - Incorrect drug interactions
- Pattern 2: Bias Issues
- Different recommendations based on demographics
 - Underrepresentation in training data
3. Root Causes (Why it happened):
- Dehumanization through algorithmic reductionism
 - Erosion of patient autonomy
 - Lack of human oversight

Problem Statement Synthesis: Combine How-What-Why into a coherent problem statement that shows the gap between ideal implementation and current reality.

4c. Acknowledging Sources (Citation)

Format: APA 7th edition

Key Rules:

- Interpret in your own words
- Cite ALL sources that informed your analysis
- Never copy exact text
- Every idea must be attributed

Example Citation:

Journal Article:
 Haltaufderheide, J., & Ranisch, R. (2024). The ethics of ChatGPT in medicine and healthcare: A systematic review on large language models. *AI & Society*, 39(2), 1-15.
<https://doi.org/10.1007/s00146-024-xxxxx>

Stage 5: Dissemination

Forms of delivery:

- Research reports
- Tools
- Recommended processes

Format depends on client requirements and intended application.

5. Types of Sources (CRITICAL TO KNOW!)

Type	Description	When to Use	Examples
Primary	Original, unfiltered materials providing direct evidence	Making claims, supporting theories, gaining direct perspectives	Interviews, original code, conference proceedings, technical reports, patents, research data
Secondary	Analyze and build upon primary sources with added interpretation	Background information, different perspectives, identifying research gaps	Textbooks, review articles, blog posts, literature reviews
Tertiary	Compiled collections presented as factual overviews	General overviews, background information, starting research	Encyclopedias, dictionaries, handbooks

Key Feature of Tertiary: Highly reliable but lack original analysis or current perspectives.

6. Workplace Applications of Research Skills

Real IT Scenarios:

1. **Project Analysis**
 - Identify problems, collect resources, evaluate quality, develop solutions
 2. **Process Improvement**
 - Research efficient workflows and new technologies
 3. **Customer Research**
 - Understand user needs, interests, pain points
 4. **Cost Optimization**
 - Identify waste, redirect resources effectively
 5. **Competitive Analysis**
 - Monitor competitors, identify opportunities
 6. **Personal Development**
 - Learn new technologies, stay current with trends
-

7. Using Generative AI Effectively

Guidelines for Ethical Use:

1. **Define your objective** - Clarify what you want to achieve
2. **Avoid plagiarism** - Never copy AI output directly
3. **Leverage for efficiency** - Use AI to gather information quickly

4. **Focus on connections** - Select relevant concepts and link them yourself

Effective Prompts (Use Action-Based Commands):

- ✓ "Summarize the key ethical issues in [source X]"
- ✓ "List the top 3 limitations of technology X"
- ✓ "Explain [concept] in simple terms with a real-world example"
- ✓ "Compare and contrast [approach A] and [approach B]"
- X "Tell me about AI ethics" (too vague)
- X "Write my assignment" (academic misconduct)

Five-Step Process for Using AI Output:

1. **Verify and Select Sources** - Check credibility, categorize as primary/secondary/tertiary
2. **Read and Identify Gaps** - Use prompts for clarification when needed
3. **Iterate** - Ask follow-up questions until you understand
4. **Analyze and Connect** - Look for patterns and relationships
5. **Synthesize** - Create your own understanding and explanations

8. Common Assessment Scenarios

Scenario 1: Source Evaluation

Question: You find an article on AI ethics. How do you evaluate it?

Answer Approach:

- Check Currency (publication date)
- Verify Accuracy (cross-reference with other sources)
- Assess Authority (author credentials, journal reputation)
- Consider Accessibility (is it peer-reviewed?)
- Evaluate Stability (is it likely to change?)

Scenario 2: Identifying Source Type

Question: Classify this source: "A textbook chapter reviewing multiple studies on machine learning bias"

Answer: Secondary source (analyzes and builds upon primary research)

Scenario 3: Research Cycle Stage

Question: You're organizing collected information by themes. Which stage are you in?

Answer: Processing of data and information

Scenario 4: Problem Statement Development

Question: Apply the How-What-Why framework to an AI ethics issue

Answer Structure:

- **How:** Describe ideal implementation
 - **What:** Identify specific ethical issues
 - **Why:** Explain root causes
 - **Integration:** Synthesize gap between ideal and reality
-

9. Quick Reference Checklist

Before Your Quiz, Can You:

- ☐ Define research and research skills?
 - ☐ List all 6 stages of the intelligent research cycle in order?
 - ☐ Distinguish between primary, secondary, and tertiary sources?
 - ☐ Apply the CAAS evaluation framework?
 - ☐ Explain the How-What-Why problem statement framework?
 - ☐ Identify appropriate workplace applications of research skills?
 - ☐ Describe ethical use of Generative AI in research?
 - ☐ Know when to use each type of source?
 - ☐ Understand the three sub-processes within Analysis?
-

10. Common Mistakes to Avoid

- ✗ **Confusing source types** - Remember: Primary = original, Secondary = analysis, Tertiary = compilation
- ✗ **Skipping source evaluation** - Always use CAAS framework
- ✗ **Copying AI output directly** - This is plagiarism; always synthesize in your own words
- ✗ **Forgetting citations** - Every idea needs attribution, even if rephrased
- ✗ **Mixing up research cycle stages** - Processing comes BEFORE analysis
- ✗ **Not identifying patterns** - Integration requires finding recurring themes across sources

Key Takeaways for Success

1. **Research is systematic, not random** - Follow the intelligent research cycle
2. **Critical thinking trumps AI confidence** - Always evaluate, don't just accept
3. **Quality over quantity** - 8-10 credible sources beat 50 poor ones
4. **Pattern recognition is crucial** - Look for How-What-Why in your sources
5. **Proper attribution is non-negotiable** - Cite everything

Remember: The goal of research skills is to transform raw information into actionable intelligence that benefits decision-making!

Week 4 Revision Notes: Meetings in IT Professional Practice

1. Meeting Fundamentals

Key Definitions

- **Meeting:** A purposeful gathering to discuss, coordinate, and accomplish shared objectives through structured collaboration
- **Two essential elements:** Structure + Communication planning (formal or informal)

Formal vs Informal Meetings

Formal Meetings:

- Pre-planned with predetermined topics
- Have formal notices, agreed agenda, chairperson
- Written minutes documenting decisions
- Examples: Board meetings, AGMs, stakeholder meetings
- **Key features:** Structured rules, controlled participation, documented outcomes

Informal Meetings:

- Unplanned and impromptu
- No formal minutes, chairperson, or predetermined agenda
- Flexible direction that can change during discussion

- **Key features:** Shared leadership, open discussion, task-oriented, adaptive
 - Common uses: Quick information exchange, collaborative problem-solving
-

2. Meetings in Software Development

Waterfall Methodology

- **Meeting structure:** Formal checkpoints at phase transitions
- Meetings concentrated at specific phases (requirements, design, testing)
- Less frequent but more formal
- **Example scenario:** Design review meeting before moving from requirements to implementation phase

Agile/Scrum Methodology

- **Meeting structure:** Continuous collaboration with regular ceremonies
- **Sprint cycle:** Typically 1-4 weeks

Core Scrum Ceremonies (Know these!):

1. **Sprint Planning** - Define sprint goals and work capacity
2. **Daily Standups** - Share progress, goals, and blockers (15 min max)
3. **Sprint Reviews** - Demonstrate completed functionality
4. **Retrospectives** - Reflect on process improvements

Application Example: In a 2-week sprint, you'd have 1 planning meeting, 10 daily standups, 1 review, and 1 retrospective.

3. Meeting Effectiveness & Best Practices

Essential Functions Across Methodologies:

- Technical decision-making
- Cross-functional collaboration
- Project planning and tracking
- Knowledge sharing

Best Practices (Apply these in scenarios):

1. **Clear Objectives** - Define purpose and expected outcomes
2. **Right Participants** - Include only necessary stakeholders
3. **Time Management** - Respect scheduled duration

4. **Action Items** - Document decisions and next steps
5. **Continuous Improvement** - Regular evaluation

Professional Impact:

- Enhanced career growth through better communication
 - Stronger team dynamics and collaboration
 - Higher project success rates
-

4. Meeting Agendas

Essential Components:

- **Item** - Topic to discuss
- **Desired Outcome** - Specific result expected
- **Priority** - High/Medium (avoid "low" - never gets discussed!)
- **Time** - Realistic allocation (+33% buffer for beginners)
- **Who** - Person responsible
- **How** - Discussion method to use

Discussion Methods (Know when to use each):

Method	When to Use	Example
Go-around	Everyone needs to contribute	Daily standup updates
Feedback	Responding to specific proposal	Reviewing design mockups
Brainstorm	Generating ideas freely	Finding solutions to a bug
Spend-a-dollar	Prioritizing options	Ranking feature requests
List items & commit	Assigning action items	End-of-meeting task delegation

Application Scenario: If discussing "Which features to include in next sprint?" use Spend-a-dollar to prioritize, then List items & commit to assign owners.

5. Meeting Minutes

Purpose:

Transform meeting conversations into trackable commitments and organizational memory

Three Critical Elements:

1. What was **decided**

2. What needs to be **achieved**
3. By what **date**

Creation Process:

1. **Pre-planning** - Use agenda as template
2. **Record-taking** - Clarify expectations
3. **Writing** - Capture decisions, accomplishments, actions (do immediately while fresh!)
4. **Sharing** - Distribute digitally
5. **Storage** - File for accountability

Why it matters: Without minutes, no recourse if actions aren't completed - may need to repeat entire meeting!

6. STAR Technique (Interview Context)

Framework:

- **Situation** - Context you dealt with
- **Task** - Specific responsibility given
- **Action** - Steps you took
- **Result** - Outcome and learning

When to Use:

- Job interviews (behavioral questions)
- CV/Resume writing
- Performance reviews
- Professional presentations

Example Application:

Question: "Tell me about a time you solved a difficult problem"

Bad answer: "I'm good at debugging and helped my team fix bugs"

Good STAR answer:

- **S:** "In my final year project, we discovered a critical bug 2 days before beta launch"
- **T:** "I needed to fix it without delaying the launch or impacting users"
- **A:** "I gathered the team, assigned debugging tasks by expertise, implemented a temporary workaround while developing the permanent fix"
- **R:** "Launched on time with no user issues, received positive feedback, and learned effective crisis management"

Best Practices:

- Keep examples short and focused
 - Use conversational tone
 - Focus on YOUR specific role and contributions
 - Be prepared for follow-up questions
-

7. AI Tools for Meetings

Pre-Meeting:

- AI scheduling assistants (Calendly AI, Motion)
- Automated agenda generation (ChatGPT, Claude)

During Meeting:

- Real-time transcription (Otter.ai, Microsoft Copilot)
- Action item extraction
- Sentiment analysis

Post-Meeting:

- Automated summaries (Fellow, Tactiq)
- Task assignment integration (Notion AI, Asana Intelligence)
- Meeting minutes generation

Scrum-Specific Tools:

- **Spinach.io** - AI Scrum Master (generates agendas, captures summaries)
- **Tactiq** - Real-time standup summaries, identifies blockers

Application Scenario: After a sprint retrospective, use AI to generate summary highlighting "what went well," "what needs improvement," and auto-create action items with owners assigned.

8. Application Scenarios for Quiz Preparation

Scenario 1: Meeting Type Selection

Question: Your team needs to quickly discuss a production bug affecting customers. What type of meeting is most appropriate?

Answer: Informal meeting - unplanned, flexible, focused on quick problem-solving without formal procedures needed for urgent issue.

Scenario 2: Discussion Method

Question: You're leading a meeting to decide which 3 of 10 proposed features to build next sprint. Which discussion method?

Answer: Spend-a-dollar - allows team to allocate 100 points across options to prioritize effectively.

Scenario 3: Agile vs Waterfall

Question: In which methodology would you have daily 15-minute standup meetings?

Answer: Agile/Scrum - continuous collaboration with daily synchronization, not Waterfall which has phase-based formal meetings.

Scenario 4: Meeting Minutes

Question: What are the three critical elements that must be in meeting minutes?

Answer:

1. What was decided
2. What needs to be achieved
3. By what date

Scenario 5: Professional Impact

Question: Why should IT professionals invest time in effective meetings?

Answer: Enhanced career growth (communication/leadership skills), better technical decisions, stronger team dynamics, higher project success rates.

Quick Check: Can You Answer These?

1. What's the difference between formal and informal meetings?
2. Name all 4 Scrum ceremonies and their purposes
3. What are the 6 components of a meeting agenda?
4. When would you use "brainstorm" vs "spend-a-dollar" discussion methods?
5. Give a STAR example from your own experience
6. Name 3 AI tools for meeting assistance and what they do
7. Why add a 33% time buffer to meeting agendas?

8. What happens if you don't write meeting minutes?
-

Common Mistakes to Avoid

- ✗ Confusing formal features (agenda, minutes, chairperson) with informal characteristics
 - ✗ Thinking Waterfall has daily standups (that's Agile!)
 - ✗ Using discussion methods randomly instead of matching to purpose
 - ✗ Forgetting the "Result" part of STAR
 - ✗ Including "low priority" items in agendas (they never get discussed!)
 - ✗ Writing minutes days later when details are forgotten
-

Final Tip: Think about meetings from three perspectives:

1. **Planning** - How to prepare (agenda, participants, time)
2. **Conducting** - How to run effectively (discussion methods, facilitation)
3. **Following up** - How to ensure accountability (minutes, action items)

Week 5 Revision Notes: Teamwork and Leadership

1. EFFECTIVE TEAM FUNDAMENTALS

What Defines an Effective Team?

Core Definition: A small group of people with complementary skills committed to a common purpose for which they hold themselves and each other accountable.

Four Key Characteristics:

1. **Deliverables** - Both individual results AND collective work products
2. **Skills** - Complementary abilities with mutual accountability
3. **Purpose** - Committed to common goals and approach
4. **Leadership** - Shared responsibility among members

Why Teams Matter

- **Enhanced Performance:** Teams outperform individuals on complex, interdependent tasks
- **Organizational Benefits:** Flexibility, efficiency, innovation through diverse skills
- **Cross-functional advantage:** Reduced handoffs and coordination overhead

Optimal Team Structure

Small Team Size Benefits:

- Easier coordination and decision-making
- More efficient meetings with better participation
- Stronger accountability and cohesion
- **Rule of thumb:** Larger teams = harder collaboration

Three Complementary Skills Required:

1. Technical expertise for task execution
2. Problem-solving and decision-making abilities
3. Interpersonal skills for collaboration and communication

Commitment and Purpose Elements

- **Shared meaningful purpose** that sets direction and aspirations
- **Specific, measurable goals** (milestones) to focus efforts
- **Small victories** from milestone achievement to maintain morale
- **Collective ownership** of both purpose and outcomes

Common Approach and Standards

Teams must agree on:

- Behavioral standards (how members interact)
- Task methodology (approach to completing work)
- Role clarity (who performs which tasks)
- Decision-making process (how choices are made)
- Schedule management (timing and deadlines)

Mutual Accountability Framework

Components:

- Shared responsibility for team goals and outcomes
- Individual clarity on personal responsibilities

- Collective ownership of team performance
- Peer accountability (members hold each other answerable)

2. TUCKMAN'S TEAM DEVELOPMENT THEORY

The Five Stages

Stage	Behaviors	Focus	Leadership Style Needed
Forming	Cautious, polite, exploring	Getting acquainted	Directive - provide clear structure and guidance
Storming	Challenging, disagreeing, conflict	Power struggles	Supportive coaching - facilitate conflict resolution
Norming	Cooperating, establishing standards	Process definition	Collaborative facilitation - help establish processes
Performing	Collaborative, anticipating needs	Goal achievement	Delegative oversight - empower and step back
Adjourning	Reflecting, separating	Closure and evaluation	Reflective leadership - celebrate and evaluate

Critical Insight

All teams go through an unproductive initial phase before becoming effective. Timeline varies significantly between teams.

The Three Critical Elements

Teams must address ALL three to become productive:

1. **Content** - What the team does (tasks, goals)
2. **Process** - How the team works (methods, procedures)
3. **Feelings** - How team members relate to each other

Problem: Most teams focus only on content, neglecting process and feelings - explaining why technically strong teams often underperform.

Common Team Challenges

Why teams fail to reach high performance:

- Many teams get stuck in **storming** (accept conflict as normal)
- Some never progress beyond **forming** (lack of clear direction)
- Incomplete **norming** leads to regression (processes not fully established)
- Teams only reach full effectiveness at **performing** stage

3. WORKING AS A TEAM IN PROJECTS

Determining Roles and Responsibilities

Process:

1. Understand project objectives, instructions, scope, and expected outcomes
2. Identify required tasks and their outputs
3. Create roles and responsibilities suitable for tasks

Common IT Team Roles:

Role	Responsibilities
Team Lead	Coordinates team, manages tasks, monitors workflows, hosts meetings, ensures process compliance
Product Owner	Defines product requirements, communicates with team, provides feature guidance, sets priorities
Team Members	Developers and designers who execute the actual work

Key Principle: Adapt roles to your specific project needs rather than following rigid structures.

Teamwork Process Cycle

Weekly/Bi-weekly Cycle:

1. **Planning** (Weekly/Bi-weekly)
 - Set clear deliverables
 - Ensure all team members understand goals
 - Focus on collaboration over competition
2. **Progress Updates** (1-3 days)
 - Team leader facilitates 15-minute check-ins
 - Identify blockers and share daily/weekly plans
 - Can be synchronous or asynchronous
3. **Progress Review** (1-3 days)
 - Product owner showcases completed work
 - Gather feedback and adjust deliverables if needed

Planning Best Practices

Expected Outcomes:

- Clear decision on weekly/bi-weekly work commitments

- Plan complete work increments
- Everyone feels confident about commitments

Top Tips:

- Focus on collaboration rather than competition
- Break work into manageable parts
- Factor in holidays and team member availability

Essential Collaboration Tools

Project Management:

- **Trello:** Visual boards for task organization and progress tracking

Documentation & Planning:

- **Notion:** All-in-one workspace for wikis, to-do lists, and project management

Communication:

- **Slack:** Team communication with channels and tool integrations

Selection Criteria: Choose tools that help resolve roadblocks, enable open communication, and support both synchronous and asynchronous collaboration.

4. FORMING GOOD TEAM DYNAMICS

Foundation Elements

Essential Components:

- Select appropriate team members
- Identify and agree upon clear objectives
- Determine roles and responsibilities
- Hold effective meetings
- Respect team members
- Manage conflict constructively
- Establish ground rules for communication and meeting schedules

Team Setup and Ground Rules

Shared Goals and Decision-Making: Define what success looks like for the team

Decision-making methods:

- Authority-based (leader decides)
- Majority vote
- Consensus
- Average member opinion

Communication and Meeting Norms:

- Choose communication tools (Slack, email, etc.)
- Set regular meeting times and formats
- Establish respect and feedback guidelines

Core Team Values

- Consider responses to others thoughtfully
- Criticize ideas, not people
- Respect absent team members
- Communicate ideas effectively
- Manage conflict constructively

Effective Meetings

- Create structured agendas
- Come prepared and pay attention
- Assign tasks with clear responsibilities
- Schedule next meetings
- Document and share meeting minutes

Common Team Problems and Solutions

Poor Teamwork Indicators:

- Lack of defined goals (both group and individual)
- Ignoring team member strengths/weaknesses
- Fear of failure stifling innovation
- Insufficient team meetings
- Disruptive personalities
- Absence of leadership
- Unclear processes

Troublesome Behaviors:

- **Individual:** Ego conflicts, self vs. group interest, poor self-awareness
- **Team:** Defensive attitudes, getting off-task, unclear roles and boundaries

Handling Difficult Behaviors

Problem	Solution
Talkative member	Use assertiveness skills
Quiet member	Apply good questioning techniques
Argumentative member	Use assertiveness and feedback skills
Free rider	Set clear ground rules from beginning
Group getting lost	Use structured agendas
Group digressing	Apply assertiveness skills
Indecisive group	Use facilitator and meeting skills
Feuding	Seek third-party mediation

Common Meeting Problems:

- **Hogging** (talking too much)
 - **Flogging** (beating issues to death)
 - **Frogging** (topic jumping)
 - **Bogging** (getting stuck)
-

5. LEADERSHIP IN IT TEAMS

Leadership Definition

- Getting cooperation from others to accomplish desired goals
- Influencing groups toward achievement of objectives

IT Leadership Principles

1. **Self-organizing teams:** Trust individuals and teams to make good decisions with autonomy
2. **Clear vision:** Provide direction while encouraging collaboration
3. **Obstacle removal:** Focus on clearing blockers rather than micromanaging

Key Leadership Qualities

1. **Learning culture:** Commit to continuous improvement
 2. **People over process:** Prioritize relationships and human factors
 3. **Adaptability:** Respond to change and listen to team input
 4. **Uncertainty comfort:** Navigate complexity through collective intelligence
 5. **Be the change:** Model the behaviors you want to see in the team
-

6. AI TOOLS FOR TEAMWORK IMPROVEMENT

Demonstrating Effective Use of AI Tools

Purpose: Improve teamwork processes through automation and enhanced collaboration

Available Tools:

- **ChatGPT Custom GPTs:** Study group coordinator agent
- **Claude Projects:** Deadline tracking assistant
- **Zapier AI:** Automated workflow agents
- **Microsoft Copilot Studio:** Team communication agent
- **Voiceflow:** Conversational scheduling agent

Team Productivity Assessment Framework

Core Question: *How effectively do our AI-enhanced processes and meetings improve team performance and output quality?*

Quantitative Indicators:

- **Time Efficiency:** Meeting duration vs. outcomes achieved
- **Output Metrics:** Deliverable completion rates, quality scores, revision cycles
- **Decision Velocity:** Time from problem identification to consensus
- **Resource Optimization:** Effective use of team expertise and tools

Qualitative Assessment:

- **Process Evolution:** Evidence of iterative improvement
- **Knowledge Integration:** Insights translating into actionable outcomes
- **Collaboration Quality:** Balanced participation, constructive conflict resolution
- **Adaptive Capacity:** Ability to modify processes based on what works

Evidence Collection Methods:

- Before/after comparisons with baseline measurements
- Team member self-assessment surveys
- External stakeholder feedback on deliverable quality
- Documentation of process modifications and rationale

APPLICATION SCENARIOS FOR QUIZ PREPARATION

Scenario 1: Team Stage Identification

Question Type: "Your team is experiencing conflict about who should lead different tasks. Some members are challenging each other's ideas frequently. Which Tuckman stage is this?"

Answer: Storming - characterized by conflict, disagreement, and power struggles. The appropriate leadership response is **supportive coaching** to facilitate conflict resolution.

Scenario 2: Team Problem Diagnosis

Question Type: "A team consistently misses deadlines, has unclear roles, and members don't know who's responsible for what. What's the primary issue?"

Answer: Lack of common approach and standards. The team needs to establish:

- Role clarity (who performs which tasks)
- Task methodology (approach to completing work)
- Decision-making process (how choices are made)
- Schedule management (timing and deadlines)

Scenario 3: Meeting Problem Solution

Question Type: "During meetings, one team member dominates the conversation while another rarely speaks. What solutions should be applied?"

Answer:

- For talkative member: Use assertiveness skills
- For quiet member: Apply good questioning techniques
- Establish ground rules for balanced participation

Scenario 4: Leadership Style Selection

Question Type: "A team has just been formed and members don't know each other well. What leadership style is most appropriate?"

Answer: Directive leadership - provide clear structure and guidance during the Forming stage when the team is cautious, polite, and exploring.

Scenario 5: Team Effectiveness Assessment

Question Type: "How would you evaluate whether a team is truly effective rather than just a working group?"

Answer: Check for all four key characteristics:

1. Deliverables - both individual AND collective work products
2. Complementary skills with mutual accountability
3. Committed to common purpose and goals
4. Shared leadership responsibility (not just one person leading)

Scenario 6: AI Tool Application

Question Type: "Your team struggles with coordinating meeting times and sending reminders. Which approach would best demonstrate effective use of AI tools?"

Answer: Create an automated workflow agent (e.g., using Zapier) that:

- Sends automated reminders
- Tracks task status
- Schedules meetings based on availability This addresses the coordination problem while freeing team members for higher-value work.

QUICK REFERENCE CHECKLIST

For Effective Teams, ensure:

- ✓ Small team size (easier coordination)
- ✓ Complementary skills across technical, problem-solving, and interpersonal domains
- ✓ Clear, shared purpose with measurable goals
- ✓ Agreed-upon processes and behavioral standards
- ✓ Mutual accountability framework

For Team Development, remember:

- ✓ All teams go through unproductive phases initially
- ✓ Match leadership style to team stage
- ✓ Address content, process, AND feelings (not just content)
- ✓ Normalize conflict during Storming
- ✓ Monitor for regression during changes

For Team Dynamics, implement:

- ✓ Clear ground rules from the beginning
- ✓ Structured agendas for meetings
- ✓ Regular planning and review cycles

- ✓ Appropriate collaboration tools
- ✓ Strategies for handling difficult behaviors

For Leadership, practice:

- ✓ Trust and autonomy for self-organizing teams
- ✓ Clear vision with collaborative approach
- ✓ Focus on removing obstacles
- ✓ Continuous learning culture
- ✓ People over rigid processes

COMMON QUIZ MISTAKES TO AVOID

1. **Confusing team stages** - Remember the sequence: Forming → Storming → Norming → Performing → Adjourning
2. **Ignoring the three critical elements** - Teams need content, process, AND feelings addressed
3. **Mismatching leadership styles to stages** - Each stage requires different leadership (directive → supportive → collaborative → delegative → reflective)
4. **Overlooking small team size benefits** - Larger teams = harder collaboration (not better)
5. **Thinking technical skills are enough** - Teams also need problem-solving AND interpersonal skills
6. **Forgetting mutual accountability** - Effective teams have both shared responsibility AND individual clarity
7. **Assuming all team problems need the same solution** - Different problems require different interventions (see the handling difficult behaviors table)
8. **Not understanding AI tool purpose** - AI tools should improve processes and free time for higher-value work, not just automate for automation's sake

Final Tip: Focus on understanding the **why** behind each concept, not just memorizing lists. The quiz will likely test your ability to apply these concepts to realistic team scenarios.

Week 6 Revision Notes: Computing Professional Code of Ethics and Ethical AI

1. COMPUTING PROFESSIONAL PRACTICE

Key Concepts:

What is a Computing Professional?

- Someone with specialized knowledge in algorithms, software development, database design, and digital systems
- **PLUS** the wisdom to use these skills responsibly
- More than "just a coder" – you're a decision-maker affecting society

Professional Practice = Technical Skills + Professional Standards + Ethical Decision-Making

Four Key Responsibilities:

Responsibility	What It Means	Example
To Employers	Keep confidential info secret; avoid conflicts of interest; work loyally	Don't sell employer's software on the side; disclose if recommending a vendor you have financial ties to
To Other Professionals	Give credit; follow coding standards; mentor juniors	Properly cite Stack Overflow solutions you adapt
To Users/Clients	Build safe, reliable systems; be honest about limitations	Hospital software must be fast BUT also protect patient privacy
To the Public	Educate about technology; contribute to policy; whistleblow when necessary	BART engineers who warned public about train safety flaws

2. ACM CODE OF ETHICS

Structure:

- **Principles:** Core values (e.g., "Be honest and trustworthy")
- **Guidelines:** How to apply principles in practice
- **Full code:** <https://www.acm.org/code-of-ethics>

Key Principles You Must Know:

ACM 1.3 – Be honest and trustworthy **ACM 1.5** – Honor property rights including copyrights
ACM 1.7 – Respect privacy of others **ACM 2.6** – Honor contracts, agreements, and responsibilities **ACM 3.5** – Protect the dignity of users and others

Application Scenario:

XYZ Corporation Case (Exam-Style Question)

Scenario: XYZ wants to secretly monitor employee web browsing. Chris recommends buying software from Robin (Chris's partner) without disclosing the relationship. Robin developed the software while working at previous company UVW.

Ethical Violations:

1. **Privacy violation** (ACM 1.7, 3.5) – Secret monitoring = digital eavesdropping
2. **Conflict of interest** (ACM 1.3) – Chris profits without disclosure
3. **IP theft** (ACM 1.5, 2.6) – Robin likely doesn't own the software (UVW does)

Critical Thinking: When analyzing cases, identify WHICH specific ACM principles are violated and WHY.

3. REAL-WORLD CASE STUDIES (Know These!)

BART Engineers Case

- **Situation:** Discovered safety flaws in San Francisco train system
- **Dilemma:** Keep job vs. protect passengers
- **Action:** Warned public after internal complaints failed (whistleblowing)
- **Result:** Fired for "disloyalty" but system was fixed, lives saved
- **Lesson:** Sometimes ethical duty to public outweighs loyalty to employer

Amazon Hiring Algorithm

- **Problem:** AI penalized resumes containing "women" (like "women's chess team")
- **Cause:** Trained on 10 years of male-dominated hiring data
- **Lesson:** Historical bias gets encoded into AI

COMPAS Criminal Justice

- **Problem:** Predicted reoffending rates with racial bias
- **Cause:** Training data reflected existing inequalities
- **Impact:** People's freedom affected by biased algorithms

Medical AI Bias

- **Problem:** Pulse oximeters less accurate for darker skin tones
- **Cause:** Calibrated primarily on lighter-skinned patients
- **Impact:** Delayed/inadequate medical treatment during COVID

4. AI & MACHINE LEARNING FUNDAMENTALS

Definitions You Must Know:

Algorithm: Step-by-step instructions for solving a problem

Artificial Intelligence (AI): Computer programs that simulate human intelligence

Machine Learning (ML): Teaching computers to learn patterns from data WITHOUT programming every rule

- **Sub-symbolic approach** to creating intelligence
- Example: Show 1000 cat/dog photos → computer learns to distinguish them

Deep Learning: Neural networks with many layers (subset of ML)

- Great for image/speech recognition

Natural Language Processing (NLP): Teaching computers to understand human language

- Evolved from rule-based to ML approaches
- Powers chatbots, translation

Generative AI: Systems that create new content (text, images, code)

- Built on NLP foundations
- Uses large language models + deep learning

How ML Works (5 Steps):

1. **Collect Data:** Photos, text, medical records
2. **Prepare Data:** Convert to numbers (breathless = [1])
3. **Train Model:** Computer finds patterns
4. **Test Accuracy:** Try on new data
5. **Deploy & Improve:** Use in real world, keep updating

ML Strengths vs Limitations:

Strengths	Limitations
Learns without constant reprogramming	Lacks explainability (black box)
Handles complex, unstructured data	Requires massive datasets
Identifies intricate patterns	Suffers from data bias
Improves accuracy over time	Time-consuming and costly

5. ETHICAL ISSUES IN AI (Critical for Quiz!)

Three Main Problem Areas:

1. Data Processing Problems

- Search engines encode gender stereotypes
- Example: Google Translate assumes "doctor" = male, "nurse" = female

2. Training Data Bias

- **"Garbage In, Garbage Out"**
- Facial recognition trained on white faces performs poorly on darker skin
- Medical AI trained on one demographic misdiagnoses others

3. Algorithmic Decision-Making

- AI makes life-changing decisions (loans, jobs, criminal justice)
- Problem: People often don't know AI was involved or how decisions were made

Root Causes (Know WHY These Happen):

1. **Professional Identity Crisis:** Developers see themselves as "just coders" not decision-makers
2. **Values Blindness:** Unaware of own biases and assumptions
3. **System Thinking Gap:** Focus only on technical requirements, ignore social context
4. **Decision-Making Deficit:** Not taught to think systematically about ethical implications

6. FIVE CORE PRINCIPLES FOR ETHICAL ALGORITHMS

Must Memorize These + Examples:

1. Human, Societal and Environmental Well-being

- AI should make life better for people, communities, and planet
- *Question to ask:* "Does this technology make the world better overall?"

2. Human-Centered Values

- Respect human rights, dignity, diversity, autonomy
- *Example:* Translation systems that respect cultural nuances
- *Question to ask:* "Does this respect people's fundamental rights and choices?"

3. Fairness and Inclusion

- Accessible to everyone; avoid unfair discrimination
- *Example:* Hiring algorithms that evaluate fairly regardless of gender/race
- *Question to ask:* "Who might be excluded or treated unfairly by this system?"

4. Privacy Protection and Security

- Protect personal information; maintain security
- *Example:* Health apps that analyze data locally vs. sending to servers
- *Question to ask:* "How is personal data being collected, used, and protected?"

5. Reliability and Safety

- Work consistently and safely as intended
- *Example:* Autonomous vehicles that fail safely when systems malfunction
- *Question to ask:* "What happens when this system makes mistakes or fails?"

7. WHEN CODES OF ETHICS AREN'T ENOUGH

ACM Code Limitations:

- ✓ Helps identify ethical problems
- ✓ Provides general principles
- ✓ Gives examples of what NOT to do
- ✗ Doesn't resolve conflicts between principles
- ✗ Doesn't guide complex technical decisions
- ✗ Can't anticipate every new challenge

Why Technology Outpaces Rules:

- ML systems make millions of unpredictable decisions
- Algorithms affect people in ways we're still discovering
- Ethical implications emerge only AFTER deployment
- Need **Ethical Reasoning Framework** (covered in workshop)

8. EXAM PREPARATION STRATEGIES

Scenario Analysis Framework:

When given an ethical scenario, use this approach:

1. **Identify stakeholders:** Who is affected?
2. **Identify ACM violations:** Which specific principles are violated?
3. **Explain WHY it's unethical:** Don't just state the principle, explain the harm
4. **Consider alternatives:** What should have been done instead?
5. **Apply core principles:** Which of the 5 principles are relevant?

Sample Question Type:

"A facial recognition system for campus security has 95% accuracy for Caucasian students but only 70% for Asian students. Analyze the ethical issues using the ACM Code and core principles."

Model Answer Structure:

- **Fairness issue** (Core Principle 3): System discriminates based on ethnicity
- **ACM 1.2 violation:** Avoid harm - Asian students face higher false positive/negative rates
- **Human-centered values** (Core Principle 2): Doesn't respect dignity and equality
- **Recommendation:** Don't deploy until accuracy is equal across all groups, or disclose limitations publicly

9. KEY TERMS GLOSSARY

Algorithmic Bias: When AI systems produce systematically unfair outcomes for certain groups

Autonomous Agents: AI systems that act independently without constant human input

Conflict of Interest: When personal interests interfere with professional judgment

Intellectual Property: Creative work owned by individuals/companies (includes software)

Non-Disclosure Agreement: Contract to keep company information confidential

Whistleblowing: Publicly exposing wrongdoing within an organization

Training Data: The examples used to teach machine learning models

Explainability: The ability to understand how an AI system reached its decision

10. FINAL CHECKLIST

Before the quiz, make sure you can:

- [] Explain the **four key responsibilities** of computing professionals
- [] Name at least **5 specific ACM Code principles** (with numbers)
- [] Analyze a case study identifying **multiple ethical violations**
- [] Describe **BART engineers case** and its significance
- [] Define **ML, AI, deep learning, NLP, generative AI**
- [] List **5 strengths** and **5 limitations** of machine learning
- [] Identify **three main areas** where ethical issues emerge in AI
- [] Recite all **five core principles** for ethical algorithms
- [] Explain **why codes of ethics have limitations** with modern AI
- [] Apply ethical reasoning to **novel scenarios** not covered in lectures

Remember: Ethics isn't about memorizing right answers—it's about developing the judgment to identify stakeholders, recognize conflicts, apply principles, and reason through dilemmas. Practice analyzing scenarios from multiple perspectives!

Week 7 Revision Notes: Ethical Reasoning Framework (Part 1)

Key Definitions to Know

Reasoning: Our unique human intellect to acquire knowledge and recognize truth.

Ethical: The values used to guide in making judgments of what is right and wrong based on true events.

Ethical Reasoning: Acquiring knowledge about true events by identifying and evaluating the events and being guided by values to form ethical decisions.

Framework: A foundation that consists of ideas, beliefs, or methods you use to deal with problems or how to solve them.

Ethical Reasoning Framework (ERF): A foundation that consists of a set of methods you use to decide what to do or how to develop an ethical algorithm and/or AI application.

The 4 Steps of ERF (Overview)

1. **Step 1:** Evaluate which of the ACM code of ethics the problem statement violates
2. **Step 2:** Create a solution that addresses the violation of the ACM code of ethics
3. **Step 3:** Make decisions about which ethical system theory (virtue ethics/deontology/utility) to ensure ethical coding; create pseudocode
4. **Step 4:** Design a mock-up of the solution using appropriate visualization tools

Note: This week focuses on Steps 1 and 2

STEP 1: Evaluate ACM Code of Ethics Violations

What You Need to Do:

- Identify which ACM Code of Ethics principles are violated in the problem statement
- Categorize the themes of ethical issues according to ACM code of conduct

How to Approach It:

1. Read the problem/case study carefully
2. Identify specific actions or situations that conflict with ACM principles
3. List the violated ACM codes with justification

Output from Step 1:

A categorized list of ACM code violations with explanations

Understanding the Problem Statement

Before applying Step 1, you must understand what constitutes a problem statement:

A clear problem statement describes:

- How things should work ideally
- What and why it is a problem

Three Key Components:

1. **How should AI/technology work ideally in the domain application?**
 - Example: How should a care robot ideally work in eldercare?
2. **What is the ethical issue (problem) caused by the AI techniques?**
 - Example: What problems arise from the robot's deception?

3. **Why is this a problem (why does the ethical issue matter)?**
 - Example: Why does violating autonomy and dignity matter?

Important: Your problem statement should provide clear reasoning and justifications supported by sources.

STEP 2: Create a Solution

What You Need to Do:

1. Refer to the output from Step 1 (the violations you identified)
2. Extract the users who will:
 - Directly use the application
 - Be indirectly impacted by it
3. Incorporate these users in the solution that addresses the ACM Code of Ethics violations

How to Approach It:

- Think about stakeholders affected by the ethical issue
- Design solutions that directly address each violation identified in Step 1
- Ensure your solution considers all user groups

Output from Step 2:

A proposed solution that addresses the ACM code violations and considers all relevant users/stakeholders

Application Example: "Today I am Paul"

This sci-fiction story serves as a case study example. When analyzing it:

For Step 1 - Ask yourself:

1. How should Paul ideally work in caring for Mildred?
2. What ethical issues has Paul violated from the ACM Code of Conduct? (list them)
3. Why is this a problem? (elaborate using ACM Code of Conduct)

For Step 2 - Consider:

- Who are the direct users? (e.g., Mildred, caregivers)
- Who is indirectly impacted? (e.g., family members, healthcare system)
- How can solutions address the violations while serving these users?

Teamwork Application Tips

For effective ERF teamwork:

1. **Collective Understanding:** All team members should understand and research together; collectively brainstorm the problem statement
2. **Collaborative Solutions:** Collectively brainstorm what the solution should look like
3. **Assign Roles:**
 - **Product Owner:** In charge of defining product solutions; gets ideas from team but is ultimately responsible for developing the product
 - **Team Leader:** Ensures all teams coordinate with each other in their collective tasks

Quiz Preparation Checklist

Can you:

- ☐ Define all key terms (reasoning, ethical, ethical reasoning, framework, ERF)?
- ☐ List all 4 steps of the ERF?
- ☐ Explain the three components of a problem statement?
- ☐ Identify ACM code violations in a given scenario?
- ☐ Categorize ethical issues according to ACM principles?
- ☐ Identify direct and indirect users/stakeholders?
- ☐ Propose solutions that address specific ACM violations?
- ☐ Explain how ERF applies to practical teamwork?

Common Mistakes to Avoid

1. **Not being specific:** Always tie violations to specific ACM codes
2. **Forgetting indirect stakeholders:** Consider all who are impacted, not just direct users
3. **Solutions that don't address violations:** Your Step 2 solution must directly respond to Step 1 findings
4. **Vague problem statements:** Be clear about what, why, and how things should work ideally

Practice Questions

1. What is the difference between "ethical" and "ethical reasoning"?

2. Why is the ERF important for computing professionals?
 3. Given a scenario where an AI system discriminates against certain users, what ACM codes might be violated?
 4. How would you identify both direct and indirect users of a healthcare AI application?
-

Remember: The ERF is a practical framework to apply ethical guidelines in coding and designing algorithms. Focus on understanding the process and being able to apply Steps 1 and 2 to real scenarios.

Week 8 Revision Notes: Ethical Theory Systems in Ethical Coding (ERF Step 3)

Learning Outcomes

By the end of this week, you should be able to:

1. Comprehend virtue ethics, utilitarianism, and deontology theories
 2. Differentiate between the theories and their influences on technology decisions
 3. Comprehend how to apply guiding principles in the ethical reasoning framework
-

Key Definition

Ethical System Theory: Structured philosophical frameworks that provide systematic approaches for determining what actions, decisions, or behaviors are morally right or wrong.

These theories offer different foundational principles and methodologies for ethical reasoning and decision-making.

The Three Ethical Theories

1. VIRTUE ETHICS

Core Concept: Character-based ethics that treats character as fundamental. It seeks to promote good character.

Key Virtues Include:

- Justice (fairness, equality)
- Honesty (truthfulness, integrity)
- Responsibility/Accountability
- Kindness/Care (empathy, compassion)
- Courage, temperance, wisdom, diligence, humility

Example: Courage gives us strength to endure hardship for something good; temperance gives us self-control to resist "too much of a good thing."

2. UTILITARIANISM

Core Concept: Consequentialist theory that determines ethical correctness based solely on foreseeable consequences by maximizing expected overall utility.

Founded by: Jeremy Bentham (promotes maximization of human welfare)

Key Principle: "Greatest good for the greatest number"

Decision Method:

1. Identify various actions that could be performed
2. Estimate benefits and harms from each action
3. Choose the action providing greatest benefits after costs are considered

Focus: Overall ethical consequences of actions

3. DEONTOLOGY

Core Concept: Duty-based ethics focused on following rules and obligations regardless of consequences.

Etymology: From Greek "deos" (duty) + "logos" (study of)

Key Principle: Some choices cannot be justified by their effects - certain actions are inherently right or wrong

Characteristics:

- Rule-based approach
- Absolute moral duties
- Actions judged by adherence to rules, not outcomes
- Example: "It is wrong to kill innocent people" or "It is wrong to tell a lie"

Applications in AI/Computing

Virtue Ethics in AI

Table 1: Virtues as Guiding Principles (Adapted from Hagendorff, 2020)

Virtue	Application in Computing	Guiding Principles
Justice	Determine how algorithms behave fairly in society	Algorithmic fairness, non-discrimination, bias mitigation, inclusion, equality, diversity
Honesty	Promote explainability and transparency; acknowledge errors	Explainability, interpretability, technological disclosure, acknowledge errors and mistakes
Responsibility/Accountability	Governance of technology creation choices; feel accountable for AI outcomes	Responsibility, liability, accountability, replicability, legality, accuracy, considering long-term consequences
Kindness	Develop sense for others' needs; avoid causing harm	Non-maleficence, security, safety, privacy, protection, precaution, beneficence, well-being, sustainability, autonomy, liberty, consent

Example Application: Blood Pressure App using Justice - The app's duty includes all genders, ethnicities, and age groups in data collection/processing to avoid discrimination or bias.

Utilitarianism in AI

Why Used: AI operates with large amounts of users' data and is a technological tool for people - aligns with achieving greater good for society.

Application Approach:

- One of most common approaches for decisions affecting large groups
- Instructs us to weigh different amounts of good and bad produced by actions
- Assessment of good and bad should be done by people using the system

Example Application: Blood Pressure App - Apply utilitarian standards by:

- Following utilitarian standard of saving maximum number of statistical lives

- Distributing risk equally to different users
- Keeping parameters anonymized (no age/gender identification)

Use Cases: Cost-benefit analysis in medical diagnostics, "big data" analytics, robots for elderly care, autonomous weapons systems

Deontology in AI

Why Used: Often combined with Asimov's Three Laws of Robotics as guiding principles

Asimov's Three Laws:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm
2. A robot must obey orders given by human beings except where such orders conflict with the First Law
3. A robot must protect its own existence as long as such protection does not conflict with First or Second Laws

Values Encompassed: Beneficence, non-maleficence, fairness, privacy protection

Example Application: Blood Pressure App using Deontology:

- Minimize harm to direct users and indirect users (doctors)
 - Minimize harm to any user as long as it doesn't conflict with first rule
 - Do not use irrelevant data as long as it doesn't conflict with previous rules
 - Follow all applicable rules as long as it doesn't conflict with previous 3 rules
-

Limitations of Each Theory

Limitations of Virtue Ethics

Challenges:

- Too subjective for algorithmic implementation
- Difficult to code abstract concepts like "wisdom" or "compassion" consistently
- Requires machines to recognize situations requiring moral action
- **Explainability problem:** Most pressing challenge - how were virtues formed through experience? How did the system reach its particular action?
- **Corner cases:** Poorly represented by training data can lead to unwanted decisions
- Cannot be regarded as moral agents but rather "weak actors of responsibility"
- **Representation of reality:** Limited; needs good universality and generalization
- **Technical feasibility:** Shown (e.g., ChaffeurNet for autonomous driving) but incomplete

Conclusion: Truly applying virtue ethics to AI seems impermissible until questions of explainability and responsibility can be answered.

Limitations of Utilitarianism

Challenges:

- **Requires complex calculations** of happiness/wellbeing that are difficult to quantify
- **May justify deception** if it reduces immediate distress
- **Difficult to weigh** long-term vs. short-term benefits in real-time
- **Technological limitations:** Cameras cannot determine "worth" of individuals
- **Scenario explosion:** More information = drastically more scenarios to evaluate
- **Less transparent:** Unrestricted optimization may be less foreseeable before inspecting underlying logic
- **No explainability:** Most machine learning uses statistical analysis with no reasoning provided
- **Central ethical question:** Is it right to actively inhibit one individual's utility to achieve greater utility for others?

Self-Driving Car Example: Cannot visually analyze specific objects to determine worth of motorcyclists; would need extensive computation to run all possible scenarios.

Limitations of Deontology

Challenges:

- **Hierarchy complexity:** Building a hierarchy of constraints (forbidden/permissible/obligatory actions) is complex
- **Ignores context:** Rule-based approaches ignore context-specific information like probability of occurrence
- **Dangerous adherence:** May undertake dangerous behaviors to adhere to strict rules
- **Limited representation:** Representation of reality only possible to limited extent
- **Lower social acceptance:** Moral decisions are context-dependent, not absolute
- **Enormous complexity:** Number of rules that can conflict arbitrarily
- **Poor universality:** A rule must cover each corner case in advance
- **Only explainability advantage:** Representation of rules with prioritization provides explainability

Conclusion: From technical and functional perspective, implementing deontic approach in AVs exhibits many complications.

ERF Step 3: Apply Ethical System Theories in Code

Overview

Purpose: Use ethical system theories as guidelines in ethical coding to ensure instructions don't lead to harm, injustice, or inequity.

The 5-Step Process for Step 3

Step 1: List required computations based on proposed solution

Step 2: Determine what kind of data the solution needs

Step 3: Determine best computing methods/algorithms (NLP, ML, etc.)

Step 4: Choose most suitable ethical theory to guide the algorithm design

Step 5: Create pseudocode based on chosen ethical theory

Practical Example: "Today I am Paul" Case Study

Step 1: Required Computations

For Transparent AI Companion Solution:

- **Patient State Assessment:** Real-time cognitive capacity evaluation
- **Ethical Interaction Management:** Determining appropriate transparency levels
- **Family Communication Facilitation:** Connecting patient with real family members
- **Privacy-Preserving Care:** Monitoring health while protecting personal data
- **Consent Management:** Tracking and validating ongoing permissions

Step 2: Data Requirements

Primary Data Sources:

- **Medical Data:** Vital signs, medication schedules, emergency alerts
- **Cognitive Data:** Speech patterns, comprehension levels, memory function
- **Behavioral Data:** Sleep patterns, mobility, emotional responses
- **Interaction Data:** Communication preferences, family recognition patterns
- **Environmental Data:** Room conditions, safety sensors, comfort settings
- **Consent Data:** Family permissions, patient preferences, legal authorizations

Data Sensitivity Classification:

- **Critical:** Life-threatening medical alerts (immediate access)
 - **Sensitive:** Personal conversations, family interactions (encrypted, logged)
 - **Behavioral:** Daily patterns, preferences (anonymized after 30 days)
 - **Administrative:** Consent logs, audit trails (permanent, immutable)
-

Step 3: Computing Methods

1. **NLP:** Real-time conversation analysis for ethical interactions
 2. **Machine Learning:** Behavioral pattern recognition
 3. **Rule-Based Systems:** Ethical decision making
 4. **Computer Vision:** Non-invasive monitoring
-

Step 4: Choose Ethical Framework

Analysis of Requirements: Need framework that can:

- Handle real-time decision-making in algorithms
- Provide clear, implementable rules for AI behavior
- Balance multiple competing values (safety, autonomy, honesty)
- Work with vulnerable populations who cannot give full informed consent

Evaluation:

Virtue Ethics:

- ✓ Good for defining AI character traits
- ✗ Too subjective for algorithmic implementation
- ✗ Difficult to code consistently

Utilitarianism:

- ✓ Can optimize for measurable outcomes
- ✗ Requires complex calculations
- ✗ May justify deception
- ✗ Difficult to weigh benefits in real-time

Deontology:

- ✓ Provides clear, hard-codeable rules

- ✓ Respects patient dignity regardless of cognitive capacity
- ✓ Creates consistent, predictable AI behavior
- ✓ Prevents ethical drift over time
- ✓ Aligns with legal and regulatory requirements

CHOSEN FRAMEWORK: Deontology (Primary) + Virtue Ethics (Secondary)

Justification:

- Algorithmic compatibility with rule-based systems
- Vulnerable population protection through absolute rules
- Legal compliance with healthcare regulations
- Technical reliability preventing harmful compromises
- Stakeholder trust through predictable behavior

Step 5: Pseudocode Example

Core Deontological Rules (Absolute Constraints):

```
RULE 1: AI_IDENTITY_DISCLOSURE = MANDATORY
RULE 2: HUMAN_IMPERSONATION = FORBIDDEN
RULE 3: CONSENT_REQUIRED = TRUE
RULE 4: PATIENT_SAFETY = PRIORITY_1
RULE 5: DATA_PRIVACY = PROTECTED
```

Virtue Ethics Character Traits (Behavioral Guidelines):

```
VIRTUE_COMPASSION: Respond with empathy and understanding
VIRTUE_HONESTY: Provide truthful, transparent communication
VIRTUE_RESPONSIBILITY: Take accountability for all actions
VIRTUE_KINDNESS: Prioritize patient emotional wellbeing
VIRTUE_WISDOM: Make decisions considering long-term impact
```

Sample Pseudocode:

```
FUNCTION AI_Companion_Interaction(patient_input, context)
BEGIN
  // DEONTOLOGICAL CHECKPOINT 1: Identity Disclosure
  IF first_interaction_of_session THEN
    EXECUTE mandatory_ai_identification()
    LOG interaction_start WITH consent_verification
  END IF

  // DEONTOLOGICAL CHECKPOINT 2: Safety First
  IF detect_emergency_situation(patient_input) THEN
    EXECUTE emergency_protocol()
    NOTIFY medical_staff IMMEDIATELY
  RETURN safety_response
```

```

END IF

// VIRTUE ETHICS: Compassionate Response
// Balancing rules with kind delivery...
END FUNCTION

```

Quiz Preparation Checklist

Can you:

- ☐ Define all three ethical theories (virtue ethics, utilitarianism, deontology)?
- ☐ Explain how each theory applies to AI/computing with examples?
- ☐ List key virtues and their computing applications?
- ☐ Explain the utilitarian decision-making process?
- ☐ Describe Asimov's Three Laws of Robotics?
- ☐ Identify limitations of each ethical theory?
- ☐ Explain why hybrid approaches (combining theories) work best?
- ☐ Describe the 5 steps for applying Step 3 of ERF?
- ☐ Choose appropriate ethical theory based on context?
- ☐ Create basic pseudocode incorporating ethical principles?

Key Comparisons

Aspect	Virtue Ethics	Utilitarianism	Deontology
Focus	Character traits	Consequences/outcomes	Rules/duties
Question Asked	"What kind of person should I be?"	"What produces best outcomes?"	"What are my duties?"
Strength in AI	Defines good character	Measurable optimization	Clear, codeable rules
Weakness in AI	Too subjective	Complex calculations	Ignores context
Best For	Long-term behavior patterns	Cost-benefit analysis	Vulnerable populations
Implementability	Difficult	Moderate	Easiest

Common Mistakes to Avoid

1. **Thinking one theory is always best:** Context matters; hybrid approaches often work best
2. **Ignoring limitations:** Every theory has significant weaknesses
3. **Forgetting stakeholders:** Vulnerable populations need special consideration
4. **Oversimplifying:** Ethical coding requires intentionality and careful thought
5. **Not justifying choices:** Always explain WHY you chose a particular ethical framework
6. **Confusing theories:** Know the core difference - character vs. consequences vs. rules

Practice Questions

1. Which ethical theory would be most appropriate for an AI system managing life-support equipment? Why?
2. How would virtue ethics approach the design of a social media content moderation algorithm differently than utilitarianism?
3. What are the main challenges in implementing utilitarianism in real-time AI decision-making?
4. Why might a hybrid approach (deontology + virtue ethics) be better than using a single theory?
5. Given a healthcare AI scenario, walk through Steps 1-5 of applying ERF Step 3.

Summary: Key Takeaways

1. **Three frameworks, different approaches:** Virtue ethics (character), Utilitarianism (outcomes), Deontology (rules)
2. **No perfect solution:** Each theory has significant limitations when applied to AI
3. **Hybrid approaches work best:** Combining deontological rules with virtue-based guidelines addresses weaknesses
4. **Context is critical:** Vulnerable populations, real-time requirements, and technical constraints influence choice
5. **Intentionality required:** Every algorithmic decision has moral implications - ethical coding requires deliberate thought
6. **Step 3 of ERF:** Systematic process from computation requirements → data needs → methods → theory choice → pseudocode

Remember: The goal isn't to find the "perfect" ethical theory, but to thoughtfully apply ethical principles that protect users, respect dignity, and create trustworthy AI systems.

Week 9 Revision Notes: Step 4 - Design User-Centered Mockups

Key Learning Outcomes to Master

1. Understand why mockups/prototypes are effective communication tools
2. Translate ethical pseudocode (Step 3) into tangible interface designs
3. Apply Figma or other tools for ethical design visualization
4. Embed ethical safeguards into user interfaces

Core Concepts

1. Mockups vs Prototypes

Mockup:

- **Static** visual representation of a product/design
- Shows realistic preview of final product
- Not fully functional but visually complete
- Used to communicate design ideas

Prototype:

- Experimental process turning ideas into **tangible forms** (paper to digital)
- Can be interactive and functional
- Used for testing and validation

Key Point: Terms often used interchangeably, but prototypes tend to be more interactive.

2. Types of Prototypes (IMPORTANT for Quiz)

Low-Fidelity	High-Fidelity
Simple, incomplete version	Close to real product
Wireframes, storyboards, diagrams	Interactive UI mockups
Tests structure and flow	Tests full user experience
Quick to create	More time-intensive
Focus on functionality	Focus on visual design + interaction

Example Application:

- **Low-fidelity wireframe** for "Today I am Paul": Simple boxes showing where AI disclosure message appears
- **High-fidelity mockup:** Fully designed consent screen with colors, typography, interactive buttons

3. When to Use Mockups/Prototypes

✅ **Software:** Visualize UI/UX before coding ✅ **Product evaluation:** Test ergonomics and aesthetics ✅ **Team communication:** Connect developers, designers, stakeholders ✅ **Pitching**

ideas: Demonstrate vision to investors  **Early problem detection:** Identify issues before development

Step 4: From Pseudocode to Prototype

Critical Framework: Every Ethical Rule → Interface Element

The Translation Process:

1. **Extract rules from Step 3 pseudocode**
2. **Identify interface touchpoints** where rule applies
3. **Design specific UI elements** that enforce the rule
4. **Consider error states** when rules are violated

Example Translation Table (Know This!)

Pseudocode Rule (Step 3)	Interface Element	User Experience
AI_IDENTITY_DISCLOSURE = MANDATORY	Persistent AI indicator badge	Always-visible robot icon
CONSENT_REQUIRED = TRUE	Multi-step consent flow	Clear opt-in checkboxes
PATIENT_SAFETY = PRIORITY_1	Emergency alert system	Red alert button
HUMAN_IMPERSONATION = FORBIDDEN	Warning dialog	"This is an AI assistant" message
DATA_PRIVACY = PROTECTED	Privacy controls	Lock icons, permission toggles

Ethical Design Patterns (High Quiz Potential)

Pattern 1: Transparent AI Disclosure

Context: User first interacts with AI system

Implementation:

- Modal dialog on first use
- Persistent indicator (e.g., robot icon)
- Clear language explaining AI capabilities/limitations

Virtue Ethics Integration: Honesty through transparent communication

Quiz Scenario Example: "Design an interface element that ensures users always know they're interacting with AI, not a human."

Pattern 2: Layered Consent

Context: Different data uses require different consent levels

Implementation:

- Progressive disclosure (show information gradually)
- Granular controls (separate toggles for different permissions)
- Cannot proceed without explicit consent

Deontological Integration: Absolute rule - no action without consent

Quiz Scenario Example: *"A family member wants to view patient conversation history. What interface elements ensure proper consent is obtained?"*

Pattern 3: Ethical Override Request

Context: Emergency situations requiring rule flexibility

Implementation:

- Multi-factor authentication
- Justification text field (required)
- Automatic logging of all overrides
- Notification to affected parties

Quiz Scenario Example: *"Medical staff needs emergency access to patient data during a crisis. Design the override flow that maintains accountability."*

Stakeholder-Specific Views (Critical Concept)

Design **different interfaces** for different users:

Primary Users (Patients - "Today I am Paul" example)

- **Simplified interface** - core needs only
- **Large, accessible controls** - elderly-friendly
- **Clear privacy indicators** - trust-building
- **Emergency access features** - safety first

Secondary Users (Family Members)

- **Consent management interface** - control what they see
- **Communication history** - review past interactions
- **Privacy boundary settings** - respect patient autonomy
- **Information sharing controls** - granular permissions

Tertiary Users (Medical Staff)

- **Professional dashboard** - comprehensive view
- **Ethical compliance monitoring** - audit trails
- **Override protocols with logging** - emergency access
- **Patient consent status visibility** - quick reference

Quiz Application: *"You must design screens for three stakeholders. How would the consent interface differ for each?"*



Prototype Fidelity Progression

When to Use Each Level:

Low-Fidelity (Wireframes):

- ☒ Test ethical flow logic
- ☒ Map ethical decision points
- ☒ Validate stakeholder pathways
- ☒ Early-stage concept testing

Medium-Fidelity (Interactive Mockups):

- ☒ Test ethical interactions and feedback
- ☒ Include error states and edge cases
- ☒ Test consent flows and overrides
- ☒ Validate virtue ethics (tone, language)

High-Fidelity (Figma Prototypes):

- ☒ Demonstrate complete ethical experience
- ☒ Full visual design with accessibility
- ☒ Interactive ethical decision flows
- ☒ Realistic content and animations

Applying to "Today I am Paul" Case Study

Key Ethical Rules to Visualize:

1. Deontological Rules:

- AI must identify itself → Persistent badge/header
- No human impersonation → Clear disclaimers
- Consent required → Multi-step flow
- Patient safety priority → Emergency button

2. Virtue Ethics Guidelines:

- Compassion → Gentle, supportive language
- Honesty → Transparent about AI limitations
- Responsibility → Clear accountability trails
- Kindness → Patient emotional wellbeing focus

Example Interface Elements:

AI Disclosure Screen (First Interaction):

 AI Companion

"Hello, I'm an AI assistant here to provide companionship and support.

I am NOT a human family member, though I can help connect you with them."

[I Understand]

Family Connection Protocol:

Patient says: "I want to talk to Paul"

AI Response Interface:

"I can help you connect with Paul. Would you like me to call him for you?"

[Yes, Call Paul] [Not Now]

Note: I'm an AI assistant, not Paul himself.

Quiz Preparation Scenarios

Scenario 1: Rule-to-Interface Translation

Question Type: *"Given this pseudocode rule: `VIRTUE_COMPASSION`: Respond with empathy, design two specific interface elements that embody this virtue."*

Answer Approach:

- **Element 1:** Gentle language in error messages ("I understand this is frustrating...")
 - **Element 2:** Supportive color scheme (soft blues/greens, not harsh reds)
 - **Element 3:** Patient-paced interactions (no time limits, "Take your time" prompts)
-

Scenario 2: Stakeholder Perspective

Question Type: *"How would a consent interface differ for the patient vs. the medical staff?"*

Answer Approach:

Patient View:

- Simple yes/no choices
- Large buttons
- Plain language explanations
- Minimal options to avoid confusion

Medical Staff View:

- Detailed permission matrix
 - Override capabilities with justification
 - Audit trail access
 - Compliance indicators
-

Scenario 3: Fidelity Selection

Question Type: *"You need to test whether users understand the AI disclosure. Which prototype fidelity level is most appropriate and why?"*

Answer Approach:

- **Low-fidelity wireframe** is best for testing information comprehension
- Can focus users on content, not visual design

- Quick to iterate based on feedback
- Cost-effective for early-stage testing

Scenario 4: Ethical Pattern Application

Question Type: *"An elderly patient with dementia might forget they're talking to an AI. Design an interface solution using ethical design patterns."*

Answer Approach:

- **Pattern Used:** Transparent AI Disclosure (persistent)
- **Implementation:**
 - Always-visible AI indicator in header
 - Periodic gentle reminders ("This is your AI companion")
 - Visual distinction from human video calls (border color, icon)
 - Simple language repeated consistently

Key Advantages of Mockups (Memorize These)

1. **Cost efficiency** - Catch expensive changes early
2. **Clear communication** - Bridge between teams and stakeholders
3. **Early problem detection** - Find issues before coding
4. **Accelerated decision-making** - Visual clarity speeds approvals
5. **User feedback** - Gather input sooner
6. **Flexibility** - Explore alternatives quickly
7. **Pitching tool** - Wow investors with vision

Common Quiz Mistakes to Avoid

✗ Confusing mockup with prototype - Know the distinction **✗ Forgetting stakeholder perspectives** - Always consider who is using the interface **✗ Missing the ethical rule connection** - Every design choice should map to Step 3 pseudocode **✗ Ignoring accessibility** - Elderly, vulnerable users need special consideration **✗ Choosing wrong fidelity level** - Match the prototype type to the testing goal **✗ Not showing virtue ethics** - Remember: language, tone, and visual choices embody virtues



Practice Questions

1. **Identify the prototype type:** *"A storyboard showing how a patient interacts with the AI companion throughout their day"* → **Low-fidelity**
 2. **Map the rule:** *"PATIENT_SAFETY = PRIORITY_1"* → Emergency alert button, medical staff notification system, 24/7 monitoring indicator
 3. **Choose fidelity:** *"Testing if the consent flow is confusing"* → **Medium-fidelity** interactive mockup (need to test actual interaction)
 4. **Design pattern:** *"Users must give consent before any data sharing"* → **Layered Consent** pattern with granular controls
 5. **Stakeholder difference:** Medical staff needs **override capabilities**; patients need **simple controls**
-



Final Exam Tips

1. **Always connect back to Step 3** - Mockups visualize pseudocode rules
 2. **Think in patterns** - Disclosure, Consent, Override
 3. **Consider the user** - Elderly patient ≠ Medical professional
 4. **Justify fidelity choices** - Low for structure, high for experience
 5. **Show both deontology AND virtue ethics** - Rules + Character traits
 6. **Use "Today I am Paul" examples** - Familiar case study performs well
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Key Takeaway

Step 4 is about making ethics visible and tangible. Every ethical rule from Step 3 must become a concrete interface element that users can see, understand, and interact with. The mockup is not just a pretty picture—it's a **functional demonstration of your ethical reasoning in action**.

Week 10: Gibb's Reflective Cycle

Overview

The Gibbs Reflective Cycle (1988) is a six-stage framework for structured reflection that helps you learn from experiences. It's widely used in healthcare, education, and professional development.

The Six Stages

1. Description

What happened?

- Provide a factual account of the situation
- Include who, what, where, when
- Stay objective - no judgments or evaluations yet
- Keep it concise but complete

2. Feelings

What were you thinking and feeling?

- Identify your emotions during the situation
- Note thoughts and reactions at the time
- Consider how feelings influenced your actions
- Be honest about both positive and negative emotions

3. Evaluation

What was good and bad about the experience?

- Analyze what worked well
- Identify what didn't work
- Consider both from your perspective and others involved
- Look at the overall outcome

4. Analysis

What sense can you make of the situation?

- Dig deeper into why things happened as they did
- Consider different perspectives
- Link to theory, research, or best practices
- Identify contributing factors
- Question assumptions

5. Conclusion

What else could you have done?

- Summarize key learning points
- Identify what you could have done differently
- Consider alternative approaches
- Acknowledge what you did well

6. Action Plan

If it arose again, what would you do?

- Create specific strategies for similar future situations
- Identify skills or knowledge you need to develop
- Set concrete goals for improvement
- Plan how to implement changes

Key Benefits

- Provides structure to reflection
- Encourages comprehensive analysis
- Promotes deeper learning from experiences
- Helps identify areas for improvement
- Supports professional development

Tips for Effective Use

- Be honest in your reflections
- Take time with each stage
- Don't skip stages - they build on each other
- Write down your reflections
- Review and revisit your reflections over time
- Apply learnings to future practice

Example Questions for Each Stage

Description:

- What was the context?
- What exactly occurred?
- Who else was involved?

Feelings:

- How did I feel before, during, and after?
- What was I worried about?
- What surprised me?

Evaluation:

- What went well and why?
- What challenges arose?
- How did others respond?

Analysis:

- Why did things unfold this way?
- What influenced the outcome?
- What knowledge/skills were helpful or lacking?

Conclusion:

- What have I learned?
- What would I change?
- What would I keep the same?

Action Plan:

- How will I handle similar situations?
- What skills do I need to develop?

- What resources can support my development?

Common Pitfalls to Avoid

- Being too descriptive without analysis
- Focusing only on negatives
- Not being honest about feelings
- Rushing through stages
- Failing to create specific action plans
- Not following up on action plans

Remember

Reflection is a skill that improves with practice. Regular use of Gibbs' cycle can enhance critical thinking, self-awareness, and professional growth.